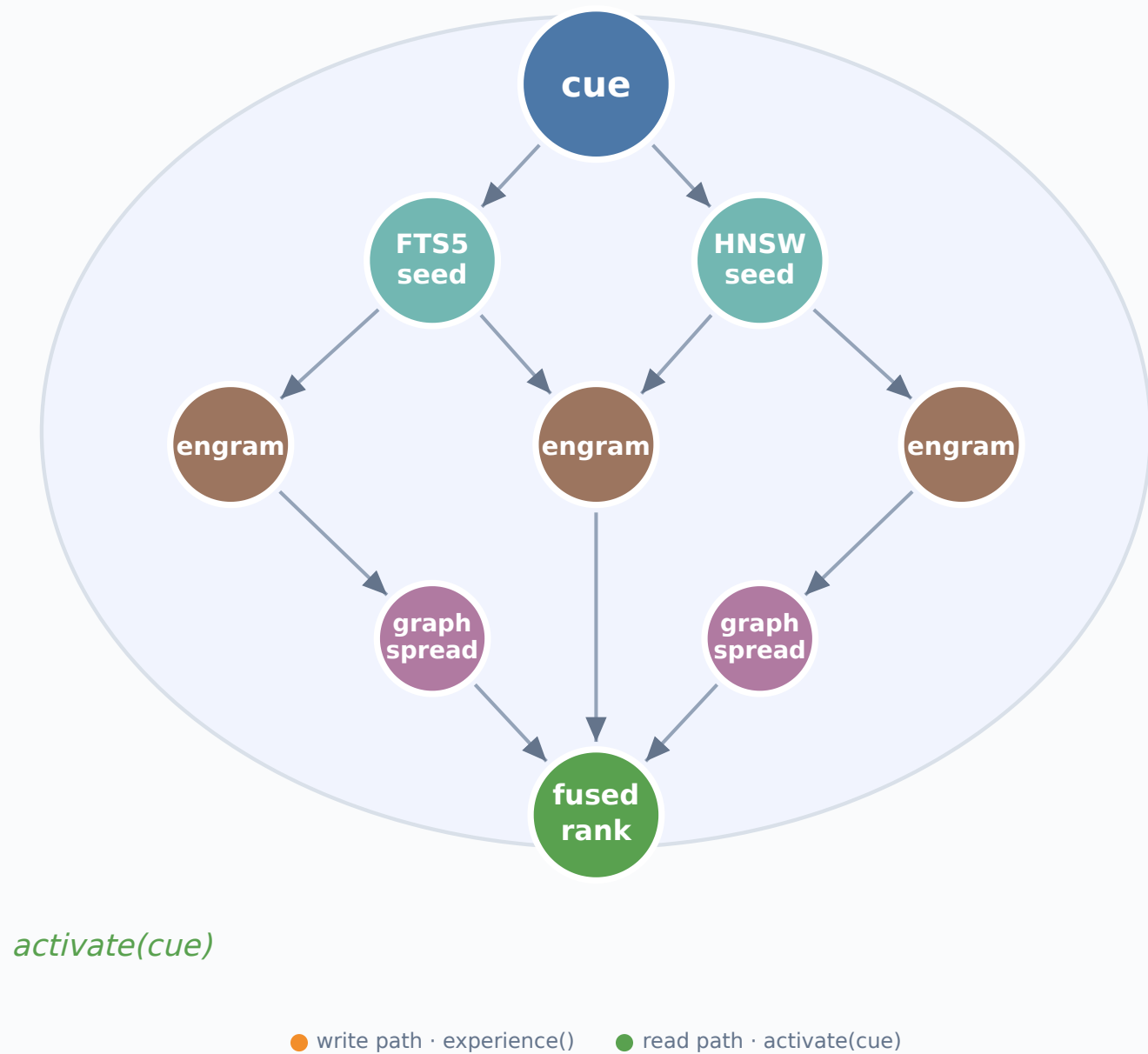


(a) Cue-driven memory activation



(b) Agent brain directory

```
manifest.json
hippocampus/      engram segments
graph/            synapse edges
recall_index.sqlite
  FTS5 + vector blobs
*.wal             → checkpoint()
```

↓ stores engram units

(c) Engram (logical unit)

```
content
.
context
.
outcome
provenance      kind, verified
rag             doc_id, chunk_id
semantic_vector optional embedding
graph wiring    co-activation edges
```

↓ upsert to sidecar

(d) Recall sidecar

FTS5 lexical seeds + HNSW semantic seeds

(e) Write: experience()

1. separation gate
2. encode episode
3. wire graph synapses
4. upsert recall sidecar

(f) Read: activate(cue)

1. lexical + semantic seed
2. graph spread (0-4 hops)
3. fuse FTS5 + vector scores
4. provenance boost → rank

(g) Third data model

Relational

- Typed rows
- Predicate match
- No graph

Vector DB

- ANN top-k
- Embedding index
- No provenance

FluctlightDB

- Engram + graph
- Hybrid FTS5+HNSW
- Agent contract